Ameya Anjarlekar

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PhD Candidate in ECE | LLM Post-Training | Trustworthy AI | Reinforcement Learning

Education

2022–present **PhD, Electrical & Computer Engineering**, *University of Illinois*, Urbana-Champaign, GPA 3.97/4.
Research Areas: Trustworthy AI, LLM Post-Training, LLM Evaluation, Reinforcement Learning, Optimization

2017–2021 : **Bachelor of Technology, Electrical Engineering (with a minor in Computer Science)**, *Indian Institute of Technology*, Bombay, CPI 9.64/10.

Relevant Publications

NeurIPS **Scalable Policy-Based RL Algorithms for POMDPs**, *NeurIPS 2025 (Main Track)*.

EMNLP **Efficient LLM Unlearning via Gradient Ratio-Based Influence Estimation and Noise Injection**, *EMNLP 2027 (Under Review)*.

Experience

05/26-present ***NVIDIA***, *Software Intern (Autonomous Vehicles)*.
- Built an observability and analysis framework for autonomous vehicle planning by instrumenting planner diagnostics and developing **SQL**-based workflows to identify passivation failure modes across driving logs
- Identified causes of unnecessary driver handovers and implemented planner logic improvements by redesigning **passivation criteria** and reducing overly conservative checks through **false-positive/false-negative** analysis

02/25-08/25 ***NVIDIA***, *Responsible AI Intern*.
- Built a modular evaluation framework for trustworthy **LLMs**, integrating **vLLM** and open-source benchmarks to evaluate privacy, robustness, bias, and broader safety risks across text, **multimodal**, and **agentic** models
- Developed a post-training **machine unlearning** framework for **7B LLMs**, implementing gradient-ratio-based influence estimation and selective noise injection to efficiently remove targeted knowledge while preserving utility
- Fine-tuned **7B Llama 2 and Zephyr** models using **Supervised Fine-Tuning** on an **8-GPU** cluster with **PyTorch** and **Hugging Face Transformers** and evaluated on the **TOFU**, **WMDP**, and **SafePKU** benchmarks
- Led to an **EMNLP** submission and a provisional **patent** for efficient LLM machine unlearning

06/21-06/22 ***Quadeye Securities***, *Quantitative Researcher*.
- Applied machine learning to quantitative pair trading strategies, improving the Sharpe ratio from 3 to 4

06/20-07/20 ***Daikin Industries***, *Machine Learning Research Scientist Intern*.
- Accomplished 50% **video data compression** by developing a Machine-learning-based image processing algorithm

Research Experience

08/24-present **Scalable Policy-Based RL Algorithms for POMDPs**.
- Developed a fast and scalable approximate **reinforcement learning** algorithm for finite-state **POMDPs**, enabling policy optimization under partial observability
- Proposed an approximate belief-state representation and combined **Temporal-Difference (TD) learning** with **Natural Policy Gradient (NPG)** to efficiently learn approximate policies in the POMDP
- Derived theoretical guarantees showing the approximation error decays exponentially with history length
- Demonstrated improved performance over existing policy-based POMDP methods on benchmark environments.

08/22-08/24 ***Mechanism Design for Heterogeneous Differentially Private Data Acquisition***.
- Introduced a mechanism design approach using ideas from game theory and statistical learning to optimize utility-privacy tradeoff in **differentially-private training** of machine learning models

08/18-12/19 **IIT Bombay Mars Rover Team**.
- Developed a **computer vision** framework for autonomous navigation, achieving 98% object detection accuracy.
- Designed a **ROS**-based perception pipeline integrating LiDAR and RGB cameras for autonomous navigation.

Technical Skills and Interests

Languages	Python, C++, C, SQL, Bash, R, Matlab
Frameworks	PyTorch, Hugging Face Transformers, vLLM, Triton, TensorFlow, Pandas, Kratos
LLM	Supervised Fine-Tuning (SFT), LoRA, Machine Unlearning, LLM Inference, RLHF
Systems	Git, Kubernetes, Slurm, CI/CD, ROS

Achievements

* Secured an All-India Rank **132** in **JEE - Advanced** and an All-India Rank **215** in **JEE - Main**